

Instrumental Variables

Session 1: Foundations and Mechanics

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Empirical Methods for PhD Students

- 1 Foundations of Instrumental Variables
- 2 2SLS and Overidentification
- 3 Instrument Strength
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Three Conditions for a Valid Instrument

A valid instrument Z_i must satisfy three conditions:

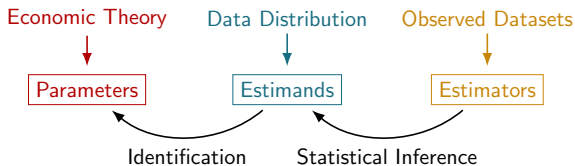
1. **Relevance (First Stage):** Z_i must be correlated with the endogenous treatment D_i
 - ▷ Testable: first-stage F-statistic (rule of thumb: $F > 10$)
2. **Independence (As-Good-As-Random Assignment):** Z_i must be uncorrelated with unobserved confounders ε_i
 - ▷ Not directly testable; requires economic reasoning and robustness checks
3. **Exclusion Restriction:** Z_i affects outcome Y_i *only* through its effect on D_i
 - ▷ No direct effect; testable indirectly via placebo/falsification

In Session 2 we add a 4th condition: **Monotonicity** (for the LATE interpretation).

Three distinct objects, not always clearly distinguished:

- **Parameters** come from economic (or other) models of the world
 - ▷ E.g. a potential-outcome model relating schooling to earnings
 - ▷ They set the *target* for empirical analyses: what we want to know
- **Estimands** are functions of the population data distribution
 - ▷ E.g. a difference in means or ratio of population regression coefficients
 - ▷ Identification: assumptions that link parameters to estimands
- **Estimators** are functions of observed (sample) data
 - ▷ E.g. a sample ratio of OLS coefficients
 - ▷ Estimators are random variables; statistical inference studies their distribution

The Lay of the Land



Separating identification (linking parameters to estimands) from inference (linking estimators to estimands) keeps the analysis disciplined.

The Endogeneity Problem

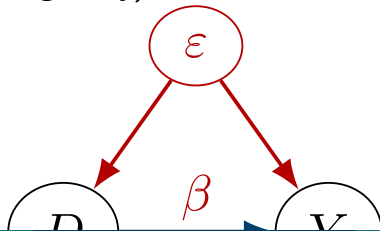
Simple causal model: $Y_i = \beta D_i + \varepsilon_i$, where D_i is the treatment, ε_i is the unobserved term.

In large samples, OLS converges to:

$$\beta^{OLS} = \beta + \frac{Cov(\varepsilon_i, D_i)}{Var(D_i)}$$

- OLS identifies β only if $Cov(\varepsilon_i, D_i) = 0$
- **Selection bias:** units with higher/lower ε_i select into treatment, so $Cov(\varepsilon_i, D_i) \neq 0$

Causal diagram (endogeneity):



The IV Solution

Suppose $Z_i \in \{0, 1\}$ is a binary instrument (e.g. lottery assignment).

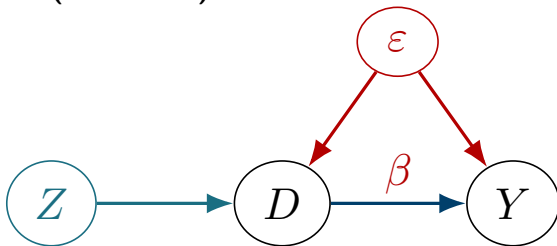
- **Randomness:** $Cov(Z_i, \varepsilon_i) = 0$ (no direct link to unobservables)
- **Exclusion:** Z_i affects Y_i only through D_i

Plugging in $\varepsilon_i = Y_i - \beta D_i$:

$$Cov(Z_i, Y_i - \beta D_i) = 0 \implies \beta = \frac{Cov(Z_i, Y_i)}{Cov(Z_i, D_i)} \equiv \beta^{IV},$$

so long as $Cov(Z_i, D_i) \neq 0$ (relevance).

Causal diagram (IV adds Z):



No arrow from Z to ε (independence) and no arrow from Z to Y

A simple decomposition links IV back to familiar OLS regressions:

$$\beta^{IV} = \frac{Cov(Z_i, Y_i)}{Cov(Z_i, D_i)} = \frac{Cov(Z_i, Y_i)/Var(Z_i)}{Cov(Z_i, D_i)/Var(Z_i)} \equiv \frac{\rho}{\pi}$$

where ρ and π come from two OLS regressions:

$$Y_i = \kappa + \rho Z_i + \nu_i \quad (\text{"reduced form"})$$

$$D_i = \mu + \pi Z_i + \eta_i \quad (\text{"first stage"})$$

- The **first stage** shows how strongly Z_i moves D_i
- The **reduced form** shows the net effect of Z_i on Y_i
- IV scales the reduced form by the first stage to recover β

If Z_i is as-good-as-random only *conditional* on covariates W_i , add them to both equations:

$$\begin{aligned}Y_i &= \kappa + \rho Z_i + W_i' \phi + \nu_i \\ D_i &= \mu + \pi Z_i + W_i' \psi + \eta_i\end{aligned}$$

- The Frisch-Waugh-Lovell theorem tells us the effective instrument is now $\tilde{Z}_i$: the residuals from regressing Z_i on W_i
- If W_i is a set of randomization-strata dummies, $\tilde{Z}_i$ captures within-strata variation in Z_i
- Key: controls should be chosen on substantive grounds, not to maximize the first-stage F-statistic

Classic Example: Angrist (1990) Vietnam Draft Lottery

Angrist (1990, AER) estimated the earnings effect of military service using Vietnam-era draft-eligibility as an instrument.

- $Z_i \in \{0, 1\}$: randomly assigned draft eligibility (lottery number)
- $D_i \in \{0, 1\}$: actual military service
- Y_i : later-life earnings

With binary Z_i , the IV estimand simplifies to the **Wald estimand**:

$$\beta^{IV} = \frac{E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]}{E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]}$$

- Numerator: earnings effect of draft eligibility (reduced form)
- Denominator: effect of eligibility on actual service probability (first stage)
- Finding: military service reduces early-career earnings by $\approx 15\%$

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When $\dim(Z_i) = L > J = \dim(X_i)$, the system is *overidentified*:

$$\text{Cov}(\tilde{Z}_i, Y_i - X_i' \beta) = 0 \implies \underbrace{\text{Cov}(\tilde{Z}_i, Y_i)}_{L \times 1} = \underbrace{\text{Cov}(\tilde{Z}_i, X_i')}_{L \times J} \underbrace{\beta}_{J \times 1}$$

- We have more moment conditions than parameters
- Any full-rank $J \times L$ matrix M gives a valid IV estimator:
 $\beta^{IV}(M) = (M \text{Cov}(\tilde{Z}_i, X_i'))^{-1} M \text{Cov}(\tilde{Z}_i, Y_i)$
- Choice of M trades off efficiency vs. interpretability

Two-Stage Least Squares (2SLS)

2SLS chooses $M' = \pi$ (the matrix of first-stage coefficients):

- Effective instrument: $\tilde{Z}_i^* = \tilde{Z}_i' \pi$ (residualized first-stage fitted values)
- Combine instruments according to their predictive power for $X_i \Rightarrow$ minimizes the 2SLS standard error

Two-stage interpretation:

1. *First stage*: regress X_i on Z_i and W_i ; save fitted values $\hat{X}_i$
2. *Second stage*: regress Y_i on $\hat{X}_i$ and W_i

Practical warning

Never run 2SLS by hand—the second-stage standard errors are wrong. Use `ivreg2` (Stata) or `AER::ivreg()` / `fixest::feols()` (R).

2SLS as a Weighted Average of Just-Identified IVs

When there is a single treatment ($J = 1$), 2SLS has an elegant weighted-average representation:

$$\beta^{2SLS} = \sum_{\ell} \omega_{\ell} \beta_{\ell}^{IV}, \quad \omega_{\ell} = \frac{\pi_{\ell}^2 \text{Var}(\tilde{Z}_{i\ell})}{\sum_{\ell'} \pi_{\ell'}^2 \text{Var}(\tilde{Z}_{i\ell'})}$$

- $\beta_{\ell}^{IV} = \rho_{\ell}/\pi_{\ell}$ is the just-identified IV estimate using instrument ℓ alone
- Weights ω_{ℓ} are proportional to each instrument's contribution to first-stage variation
- Instruments with stronger first stages receive larger weights
- This “visual IV” representation (slope of ρ_{ℓ} on π_{ℓ}) is also useful for diagnosis

Under the *constant-effects* model, all just-identified IVs identify the same β :

- If β_ℓ^{IV} differ significantly across instruments, something is wrong
- The Sargan–Hansen statistic tests $H_0 : \beta_\ell^{IV} = \beta$ for all ℓ ; asymptotically $\chi^2(L - J)$ under the null

Cautions:

- Low power when individual $\hat{\beta}_\ell^{IV}$ are noisy
- Rejection does *not* tell you which instrument is invalid
- Under *heterogeneous* effects (Session 2), two valid IVs can legitimately have different estimands; the test is not interpretable
- Use the test as one input among many, not as a pass/fail criterion

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If π is small relative to its standard error, the instrument is “weak”:

- Rule of thumb: first-stage $F < 10$ signals potential weakness (Staiger and Stock, 1997)
- **Bias**: 2SLS is biased toward OLS in small samples; the bias is proportional to $1/F$ approximately
- **Size distortion**: standard confidence intervals may under-cover in finite samples

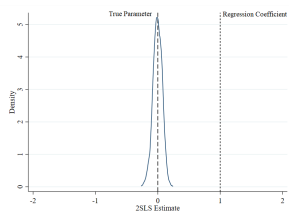
Modern guidance (Angrist and Kolesár 2022):

- In the just-identified case, 2SLS is *median-unbiased*; the SE increase tends to “cover up” the bias
- Use `weakivtest` (Stata; Montiel Olea and Pflueger 2015) for more nuanced critical values
- If F is small: consider Anderson–Rubin or JIVE as robust alternatives

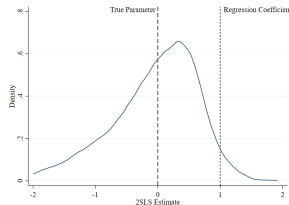
Weak Instruments: Monte Carlo

$$Y_i = 0 \cdot D_i + \varepsilon_i, D_i = \pi Z_i + \eta_i, N = 100 \text{ replications}$$

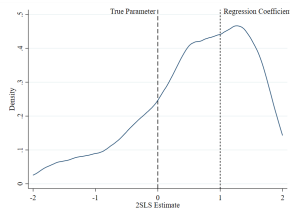
Strong ($\pi = 1, F \approx 100$)



Medium ($\pi = 0.1, F \approx 10$)



Weak ($\pi = 0.01, F \approx 1$)



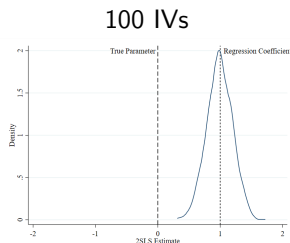
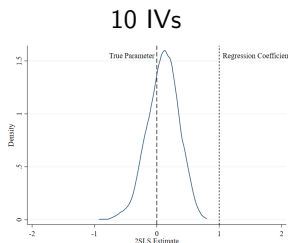
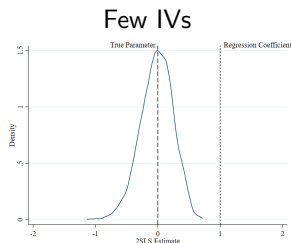
Distribution of $\hat{\beta}^{IV}$ widens dramatically as the instrument weakens. With a very weak instrument, $\hat{\beta}^{IV}$ converges to $\hat{\beta}^{OLS}$.

Many-IV Bias

Overidentified 2SLS faces an additional problem when there are many weak instruments:

- **Symptom:** low first-stage F even though SEs are small (more IVs $\Rightarrow$ richer first stage $\Rightarrow$ better fit of D_i)
- **Mechanism:** overfitting the first stage essentially recreates the endogenous variation in D_i , biasing 2SLS toward OLS
- **Historical case:** Angrist–Krueger (1991) QoB IV interacted with state $\times$ year fixed effects created hundreds of instruments

Visualized — 100 replications:



Prevention:

- Prefer parsimonious instrument sets; avoid mechanical many-IV settings
- If using a Bartik/shift-share IV with many industries, the *effective* number of instruments is the number of industries (addressed in Session 2)

Diagnosis:

- Report first-stage F-statistic (or Kleibergen–Paap for clustered SEs)
- Use `weakivtest` in Stata for heteroskedasticity-robust critical values
- See also Lee et al. (2020) for *t*-ratio-based inference

If the instrument is weak:

- Try a more informative functional form for Z_i
- Collapse to fewer instruments
- Use Anderson–Rubin confidence sets (valid regardless of strength)

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Overview: Two Landmark IV Papers (Post-2010, Top-5 Journals)

We study two highly cited papers that use **non-shift-share** IV designs to establish causal relationships:

1. **Maestas, Mullen, and Strand (2013, AER)**

“Does Disability Insurance Receipt Discourage Work?”

IV: randomly assigned Social Security disability examiners. Context: US disability insurance applicants.

2. **Aizer and Doyle (2015, QJE)**

“Juvenile Incarceration, Human Capital, and Future Crime”

IV: randomly assigned juvenile court judges. Context: US juvenile criminal justice system.

Both papers use **random assignment of decision-makers** as the identifying instrument—a design that is transparent, credible, and widely applicable in policy evaluation contexts.

Citation

Maestas, N., Mullen, K.J., and Strand, A. (2013). “Does Disability Insurance Receipt Discourage Work? Using Examiner Assignment to Estimate Causal Effects of SSDI Receipt.” *American Economic Review*, 103(5): 1797–1829. (≈1,300 Google Scholar citations)

Research question: Does receiving Social Security Disability Insurance (SSDI) benefits *cause* a reduction in employment?

Context & motivation:

- SSDI provides income support to workers with disabling conditions; by 2010 over 8.2 million Americans received benefits (\$119 billion)
- Key policy concern: does SSDI create a “poverty trap” by discouraging recipients from returning to work?
- OLS is severely biased: healthier rejected applicants would work anyway; sicker accepted applicants could not work regardless (*reverse causality + selection on unobserved health*)

The IV: Randomly assigned disability examiners differ systematically in their allowance rates (“leniency”):

- SSDI applications are assigned to examiners quasi-randomly within local Disability Determination Services (DDS) offices
- Examiners vary substantially in their allowance rates even after controlling for case characteristics
- An examiner's average grant rate (leave-one-out, in other cases) is used as the instrument for the applicant's SSDI receipt

IV validity:

- **Relevance:** examiner leniency strongly predicts own award probability (first-stage $F > 100$)
- **Independence:** assignment is quasi-random within offices (applicants cannot choose their examiner)
- **Exclusion:** examiner leniency affects employment only through SSDI award, not through other channels
- **Monotonicity:** a stricter examiner can only reduce (never increase)

Data:

- Social Security Administration administrative records: ≈ 1.5 million SSDI applicants, 1997–2004
- **Treatment:** binary indicator for SSDI award (0/1)
- **Outcomes:** employment indicator, earnings (linked to earnings records from IRS W-2 forms)
- **Instrument:** leave-one-out examiner allowance rate (average grant rate of the assigned examiner on all other cases)
- **Strata:** DDS office $\times$ quarter fixed effects (controls for within-office assignment patterns)

Specification:

$$Y_{it} = \alpha + \beta \text{SSDI}_i + W_i' \gamma + \varepsilon_i$$

- 2SLS using examiner leniency as Z_i , conditional on DDS-quarter FEs
- Robust to controlling for applicant characteristics (age, diagnosis, prior earnings) because assignment is as good as random

Key findings:

- SSDI receipt reduces employment by $\approx$ **28 percentage points** at ages 45–64 (IV estimate much larger than OLS)
- About **23%** of marginal SSDI recipients (compliers) could work: the “work-capable” group whose decision is affected by examiner assignment
- OLS understates the causal effect because rejected applicants who remain employed create a spurious positive “health effect”

Robustness and sensitivity:

- *Placebo test*: examiner leniency does not predict pre-application employment trends (balance on pre-period outcomes)
- *Alternative examiner measures*: using quintiles of leniency instead of the continuous measure gives nearly identical results
- *Sub-group consistency*: effects are similar across age groups, diagnosis categories, and states
- *Complier analysis*: marginal recipients are younger, healthier, and

Academic contributions:

- First credible causal estimate of SSDI's effect on employment; resolves a long-standing identification debate
- Shows SSDI creates work disincentives for a sizeable share of recipients—important for disability insurance design
- Introduced the “examiner leniency” IV into economics; spawned a large literature on disability insurance, screening, and physician/administrative judgment

IV methodology lessons:

- **Random assignment of decision-makers** is a powerful source of exogenous variation when direct randomization is infeasible
- The **leave-one-out leniency measure** avoids mechanical correlation between one's own case and the instrument
- **LATE interpretation** is transparent: IV estimates the effect of SSDI on “compliers”—those whose award decision is influenced by examiner assignment

Citation

Aizer, A. and Doyle, J.J., Jr. (2015). "Juvenile Incarceration, Human Capital, and Future Crime: Evidence from Randomly Assigned Judges." *Quarterly Journal of Economics*, 130(2): 759–803. ($\approx$ 900 Google Scholar citations)

Research question: Does incarcerating juveniles reduce or increase their subsequent criminal activity and human capital?

Context & motivation:

- The US incarcerates approximately 70,000 juveniles on any given day; recidivism rates are high ($\approx$ 70% re-arrested within 3 years)
- Policy debate: does incarceration deter future crime (deterrence), or does it increase it via criminal networks and disrupted schooling?
- OLS is biased: more serious offenders are both more likely to be incarcerated *and* more likely to re-offend regardless

The IV: Randomly assigned juvenile court judges vary in their propensity to incarcerate:

- Juvenile cases are assigned to judges *randomly* within courthouses in Chicago, Illinois
- Judges differ substantially in their incarceration rates even for similar offense types and defendant characteristics
- A judge's leave-one-out incarceration rate on other cases is the instrument for whether defendant i is incarcerated

IV validity:

- **Relevance:** judge leniency strongly predicts incarceration (first-stage $F \approx 35\text{--}50$)
- **Independence:** assignment is explicitly random within court (confirmed by balance tests on pre-determined characteristics)
- **Exclusion:** judge assignment affects outcomes only through the incarceration decision, not through other channels
- **Monotonicity:** a stricter judge cannot reduce a defendant's

Data:

- Illinois Juvenile Court data linked to adult criminal records, high school graduation records, and mortality data; 1990–2006
- $\approx 35,000$ juvenile cases in Chicago (ages 10–17)
- **Treatment:** binary incarceration indicator (0/1)
- **Outcomes:** adult incarceration, high school graduation, employment, and earnings in adulthood (ages 25–35)
- **Instrument:** leave-one-out judge incarceration rate (within courthouse, controlling for offense type and year)

Specification:

$$Y_i = \alpha + \beta \text{Incarcerated}_i + W_i' \gamma + \varepsilon_i$$

- 2SLS using judge leniency as Z_i , with courthouse-year-offense category fixed effects as strata
- Outcomes measured 5–15 years after the juvenile offense

Key findings:

- Juvenile incarceration *reduces* high school graduation by **≈ 13 percentage points**
- Juvenile incarceration *increases* adult incarceration probability by **$\approx 22\text{--}26$ percentage points**
- Earnings are reduced significantly; effects persist for decades
- Net cost-benefit analysis: incarceration costs society roughly \$1.1–1.4 million per juvenile (present value), far exceeding any deterrence benefit

Robustness and sensitivity:

- *Balance test*: judge leniency is uncorrelated with defendant race, prior offenses, and offense severity (random assignment holds)
- *Placebo*: judge leniency does not predict pre-offense outcomes (no pre-existing differences across complier groups)
- *Different charge types*: results hold across assault, drug, property, and public order offenses

Academic contributions:

- First causal evidence that juvenile incarceration *increases* adult criminality and reduces human capital—a critical finding for criminal justice reform debates
- Demonstrates that the criminal justice system can perpetuate rather than reduce criminal behavior through incapacitation effects
- Extended the judge-leniency IV approach (Kling 2006) to juvenile settings and linked to long-run human capital outcomes

IV methodology lessons:

- **Judge/examiner leniency IV** is now a standard tool whenever administrative decisions are quasi-randomly assigned
- Key to validity: *random* (not quasi-random) assignment within strata is what makes the exclusion credible
- **LATE**: IV identifies the effect among “compliers”—juveniles whose incarceration decision is swayed by judge assignment, typically those near the incarceration margin

Comparing the Two Papers

	MMS (2013)	AD (2015)
Journal	AER	QJE
Setting	US SSDI applicants	US juvenile defendants
Treatment	SSDI benefit receipt	Juvenile incarceration
IV type	Examiner leniency	Judge leniency
Assignment	Quasi-random (within DDS office)	Random (within courthouse)
Unit	Individual applicant	Individual defendant
Key finding	SSDI ↓ employment 28pp	Incarceration ↓ graduation 13p
Citations	≈1,300	≈900

Common thread: Both papers exploit **random variation in decision-maker leniency** to identify causal effects of high-stakes policy interventions. The IV design is clean, credible, and transparent — a template for policy evaluation research.

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IV Foundations:

- IV validity requires relevance, independence, and exclusion
- $\beta^{IV} = \rho/\pi$ (reduced form $\div$ first stage)
- With controls: effective instrument $\tilde{Z}_i$ = residual of Z_i on W_i (Frisch–Waugh)

2SLS and Overidentification:

- 2SLS = efficiency-optimal combination of multiple instruments
- 2SLS = weighted average of just-identified IVs
- Sargan–Hansen test: diagnostic, not a “certification of validity”

Instrument Strength:

- $F > 10$ rule of thumb; `weakivtest` for robust critical values
- Many-IV bias: overfitting first stage recreates endogeneity

Applications (Top-5, Post-2010):

- Maestas, Mullen & Strand (2013, AER): examiner leniency IV $\Rightarrow$ SSDI receipt reduces employment by 28pp

1. IV Interpretation (LATE)

- ▶ Imbens–Angrist (1994) assumptions: monotonicity, compliers
- ▶ Characterizing compliers; diff-in-diff IV

2. Formula Instruments

- ▶ Shift-share IV: formal conditions (Borusyak et al. 2022)
- ▶ Recentered IV: beyond shift-share (Borusyak and Hull 2023)
- ▶ Illustration: China High-Speed Rail market access

3. IV with Chinese Economic Context

- ▶ Meng, Qian, Yared (2015, *ReStud*): Great Famine IV
- ▶ Khandelwal, Schott, Wei (2013, *QJE*): WTO quota removal

Instrumental Variables — Session 1

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Questions welcome via email.

Session 2 slides cover IV interpretation and Chinese applications.